



9214 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Charlotte Mason (PI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Prof. Daniel P. Stark (CoI) (CoPI) (US Admin CoI)	University of California - Berkeley
Dr. Michael Topping (CoI)	University of Arizona
Dr. Zuyi Chen (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Mengtao Tang (CoI)	Shanghai Jiao Tong University
Dr. Viola Gelli (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Peter Senchyna (CoI)	Carnegie Institution of Washington
Dr. Ryan Endsley (CoI)	University of Texas at Austin

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	2	GOODSN_P1	NIRSpec MultiObject Spectroscopy	(10) goodsn_apr_v2p1p2
	10	GOODSN_replanP2A	NIRSpec MultiObject Spectroscopy	(11) goodsn_apr_replan_v2p4
	7	GOODSS_P1	NIRSpec MultiObject Spectroscopy	(12) goodss_aprP1_v4p1
	8	EGS_replanP4	NIRSpec MultiObject Spectroscopy	(7) egs_apr_P4_default
	9	A2744_replan	NIRSpec MultiObject Spectroscopy	(6) A2744_v4

ABSTRACT

JWST is transforming our view of star formation and supermassive black hole growth in the first billion years. NIRCам imaging has revealed an unexpected abundance of bright $z>9$ galaxy candidates. NIRSpec prism spectroscopy has shown several of the brightest sources show extremely high

ionization emission lines and peculiar abundance patterns, implying the presence of very massive stars and/or AGN, and detections of Lyman-alpha emission at $z > 10$ suggest these hard ionizing sources may drive an earlier start to reionization than we had expected. However, our understanding has been limited to sources with the brightest rest-frame UV emission lines.

The critical next step is ultra-deep high resolution rest-frame UV spectroscopy of galaxies at our redshift frontier to: confidently distinguish between massive stars and AGN powering galaxies' ionizing spectra, reveal direct emission from very massive stars, uncover the growth mode of high-redshift AGN, measure the physical conditions which facilitate early star formation and black hole growth, and constrain the onset of reionization.

We propose SPURS: the SPectroscopic Ultra-deep Reionization-era Survey - 160 hrs of R~1000 NIRSpec spectroscopy of >150 $z \sim 5-14$ galaxies, selected primarily from the deepest public spectroscopic surveys, including the majority of spectroscopically confirmed $z > 9$ galaxies, with 30 hr depths in G140M. SPURS will provide the community with a vital legacy dataset for understanding the properties of the first galaxies and AGN, and what drove the reionization process.

OBSERVING DESCRIPTION

We will target 4 legacy fields with ultra-deep spectroscopy.

Our science goal requires the R~1000 gratings, with 30 hrs of exposure in G140M/F100LP, 8 hrs of exposure in G235M/F170LP and 3 hrs of exposure in G395M/F290LP. In order to create MOS configurations, we compiled a catalog of spectroscopically selected sources from existing JWST/NIRCam grism and NIRSpec prism spectroscopy of the field.

We use the MSA planning tool to design the MOS configurations and create the observations. For dithering, we choose to nod in the 3-shutter slitlet, which is appropriate for our spatially compact high redshift galaxy targets. This approach has less overhead and maximizes multiplexing efficiency, compared to the fixed or flexible dithering options that require additional MSA configurations. At the time of flight-ready program submission, we will include extra background shutters in empty parts of the MSA, in order to ensure reliable background subtraction.

We set up our NIRSpec exposures using NRSIRS2, which is recommended for deep observations.

Proposal 9214 - Targets - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GS_catalog	RA: 03 32 32.0208 (53.1334200d) Dec: -27 48 15.16 (-27.80421d) Equinox: J2000		
	Comments: Description= []				
	(2)	GN_catalog	RA: 12 36 45.9480 (189.1914500d) Dec: +62 13 42.86 (62.22857d) Equinox: J2000		
	Comments: Description= []				
	(6)	A2744_v4	RA: 00 14 22.3021 (3.5929254d) Dec: -30 24 9.80 (-30.40272d) Equinox: J2000		
	Comments: Description= []				
	(7)	egs_apr_P4_default	RA: 14 19 58.1952 (214.9924800d) Dec: +52 54 18.42 (52.90512d) Equinox: J2000		
	Comments: Description= []				
	(10)	goodsn_apr_v2p1p2	RA: 12 36 39.3661 (189.1640254d) Dec: +62 14 42.72 (62.24520d) Equinox: J2000		
Comments: Description= []					
	(11)	goodsn_apr_replan_v2p4	RA: 12 36 34.8906 (189.1453775d) Dec: +62 14 25.11 (62.24031d) Equinox: J2000		
Comments: Description= []					
	(12)	goodss_aprP1_v4p1	RA: 03 32 39.1572 (53.1631550d) Dec: -27 47 5.63 (-27.78490d) Equinox: J2000		
Comments: Description= []					

Proposal 9214 - Observation 2 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Observation	Proposal 9214, Observation 2: GOODS_N_P1											Tue Jun 16 21:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#1) has 1 filler slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#1) has 1 primary slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#2) has 1 filler slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#2) has 1 primary slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#3) has 1 filler slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#3) has 1 primary slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#4) has 1 filler slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#4) has 1 primary slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#5) has 1 filler slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#5) has 1 primary slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#6) has 1 filler slits affected by failed closed shutters.											
	(GOODSN_P1 (Obs 2)) Warning (Form): Config c1 (#6) has 1 primary slits affected by failed closed shutters.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(10)	goodsn_apr_v2p1p2	RA: 12 36 39.3661 (189.1640254d) Dec: +62 14 42.72 (62.24520d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153		
	2	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	targets (1764 sources)	targets (1764 sources)	jwst-nirspec-mr		1.5			

Proposal 9214 - Observation 2 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	6013	189.126886	62.230309	25.0	1	6165	189.015341	62.244995	23.67	
	1	6019	189.055035	62.257253	24.41	1	6183	189.042068	62.257102	23.95	
	1	6025	189.130308	62.241338	24.23	1	6190	189.047434	62.257864	24.75	
	1	6059	189.061938	62.255616	24.84	1	6378	189.063663	62.212367	24.71	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	6013	189.126886	62.230309	25.0	2	6165	189.015341	62.244995	23.67	
2	6019	189.055035	62.257253	24.41	2	6183	189.042068	62.257102	23.95		
2	6025	189.130308	62.241338	24.23	2	6190	189.047434	62.257864	24.75		
2	6059	189.061938	62.255616	24.84	2	6378	189.063663	62.212367	24.71		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.07076208333 33 Degrees 62.234186111111 114 Degrees	316.51740513217 12			3	18	26522.602
	2	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.07076208333 33 Degrees 62.234186111111 114 Degrees	316.51740513217 12			3	18	26522.602
	3	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.07076208333 33 Degrees 62.234186111111 114 Degrees	316.51740513217 12			3	18	26522.602
	4	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.07076208333 33 Degrees 62.234186111111 114 Degrees	316.51740513217 12			3	18	26522.602
	5	2 (G235M/F170LP)	c1	3 Shutter Slitlet	189.07076208333 33 Degrees 62.234186111111 114 Degrees	316.51740513217 12			3	30	28886.002
	6	3 (G395M/F290LP)	c1	3 Shutter Slitlet	189.07076208333 33 Degrees 62.234186111111 114 Degrees	316.51740513217 12			3	9	10635.301
Special Requirements	Group Visits within 53.0 Days Visits Same PA MSA Scheduled Aperture PA 316.5998 to 316.5998 Degrees (V3 178.02527 to 178.02527)										

Proposal 9214 - Observation 10 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Observation	Proposal 9214, Observation 10: GOODSN_replanP2A										Tue Jun 16 21:00:52 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 10:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(11)	goodsn_apr_replan_v2p4	RA: 12 36 34.8906 (189.1453775d) Dec: +62 14 25.11 (62.24031d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: CLEAR; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2	Filter: CLEAR; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	fixed (76 sources)	fixed (76 sources)	jwst-nirspec-mr	1.5			
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	6165	189.015341	62.244995	23.67	1	6417	189.028600	62.203070	21.7	
	1	6195	189.100499	62.253305	22.38	1	6422	189.024871	62.205487	22.28	
	1	6207	189.005861	62.242988	21.91	1	6428	189.039815	62.209127	23.57	
	1	6209	189.076443	62.240683	22.00	1	6431	189.029633	62.210718	23.18	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	6165	189.015341	62.244995	23.67	2	6417	189.028600	62.203070	21.7	
	2	6195	189.100499	62.253305	22.38	2	6422	189.024871	62.205487	22.28	
	2	6207	189.005861	62.242988	21.91	2	6428	189.039815	62.209127	23.57	
	2	6209	189.076443	62.240683	22.00	2	6431	189.029633	62.210718	23.18	

Proposal 9214 - Observation 10 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.06079 Degrees 62.22741111111111 11 Degrees	276.45934321082 07			3	18	26522.602
	2	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.06079 Degrees 62.22741111111111 11 Degrees	276.45934321082 07			3	18	26522.602
	3	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.06079 Degrees 62.22741111111111 11 Degrees	276.45934321082 07			3	18	26522.602
	4	1 (G140M/F100LP)	c1	3 Shutter Slitlet	189.06079 Degrees 62.22741111111111 11 Degrees	276.45934321082 07			3	18	26522.602
	5	2 (G235M/F170LP)	c1	3 Shutter Slitlet	189.06079 Degrees 62.22741111111111 11 Degrees	276.45934321082 07			3	30	28886.002
	6	3 (G395M/F290LP)	c1	3 Shutter Slitlet	189.06079 Degrees 62.22741111111111 11 Degrees	276.45934321082 07			3	9	10635.301
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 276.5341177 to 276.5341177 Degrees (V3 137.959548 to 137.959548) [MSA Selected] Visits Same PA MSA Scheduled Aperture PA 276.5341 to 276.5341 Degrees (V3 137.959548 to 137.959548)										

Proposal 9214 - Observation 7 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Observation	Proposal 9214, Observation 7: GOODSS_P1										Tue Jun 16 21:00:52 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 7:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(12)	goodss_aptp1_v4p1	RA: 03 32 39.1572 (53.1631550d) Dec: -27 47 5.63 (-27.78490d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: CLEAR; Readout: NRSRAPIDD1; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD1	3	1	4	257.682	
	2	Filter: CLEAR; Readout: NRSRAPIDD1; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD1	3	1	4	257.682	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	targets (6254 sources)	targets (6254 sources)	jwst-nirspec-mr	1.5			
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	10024	53.158414	-27.819085	24.17	1	10171	53.135754	-27.785002	23.13	
	1	10058	53.150278	-27.752331	23.73	1	10173	53.123793	-27.783494	24.46	
	1	10113	53.140332	-27.797597	22.5	1	10174	53.170711	-27.782044	23.05	
	1	10166	53.134494	-27.797632	24.48	1	10181	53.158468	-27.774048	22.92	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	10024	53.158414	-27.819085	24.17	2	10171	53.135754	-27.785002	23.13	
	2	10058	53.150278	-27.752331	23.73	2	10173	53.123793	-27.783494	24.46	
	2	10113	53.140332	-27.797597	22.5	2	10174	53.170711	-27.782044	23.05	
	2	10166	53.134494	-27.797632	24.48	2	10181	53.158468	-27.774048	22.92	

Proposal 9214 - Observation 7 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	53.159366708333 33 Degrees - 27.786678888888 9 Degrees	42.433202699042 59			3	18	26522.602
	2	1 (G140M/F100LP)	c1	3 Shutter Slitlet	53.159366708333 33 Degrees - 27.786678888888 9 Degrees	42.433202699042 59			3	18	26522.602
	3	1 (G140M/F100LP)	c1	3 Shutter Slitlet	53.159366708333 33 Degrees - 27.786678888888 9 Degrees	42.433202699042 59			3	18	26522.602
	4	1 (G140M/F100LP)	c1	3 Shutter Slitlet	53.159366708333 33 Degrees - 27.786678888888 9 Degrees	42.433202699042 59			3	18	26522.602
	5	2 (G235M/F170LP)	c1	3 Shutter Slitlet	53.159366708333 33 Degrees - 27.786678888888 9 Degrees	42.433202699042 59			3	30	28886.002
	6	3 (G395M/F290LP)	c1	3 Shutter Slitlet	53.159366708333 33 Degrees - 27.786678888888 9 Degrees	42.433202699042 59			3	12	13305.068
Special Requirements	Group Visits within 53.0 Days Visits Same PA MSA Scheduled Aperture PA 42.4314 to 42.4314 Degrees (V3 263.85687 to 263.85687)										

Proposal 9214 - Observation 8 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Observation	Proposal 9214, Observation 8: EGS_replanP4											Tue Jun 16 21:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#1) has 1 filler slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#1) has 1 primary slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#2) has 1 filler slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#2) has 1 primary slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#3) has 1 filler slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#3) has 1 primary slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#4) has 1 filler slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#4) has 1 primary slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#5) has 1 filler slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#5) has 1 primary slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#6) has 1 filler slits affected by failed closed shutters.											
	(EGS_replanP4 (Obs 8)) Warning (Form): Config c1 (#6) has 1 primary slits affected by failed closed shutters.											
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
(Visit 8:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(7)	egs_apt_P4_default	RA: 14 19 58.1952 (214.9924800d) Dec: +52 54 18.42 (52.90512d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: CLEAR; Readout: NRSRAPIDD2; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD2	3	1	4	343.577		
	2	Filter: CLEAR; Readout: NRSRAPIDD2; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD2	3	1	4	343.577		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold				
	MSATA	false	No	MSA Center	fixed (60 sources)	fixed (60 sources)	jwst-nirspec-mr	1.5				

Proposal 9214 - Observation 8 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	9020	215.012170	52.904493	24.23	1	9096	214.973209	52.944727	22.69	
	1	9025	215.037744	52.913979	24.41	1	9097	214.979175	52.949327	24.22	
	1	9063	214.970329	52.914055	24.27	1	9137	214.990387	52.916955	23.96	
	1	9090	215.008556	52.948859	24.72	1	9142	214.978554	52.921541	24.03	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	9020	215.012170	52.904493	24.23	2	9096	214.973209	52.944727	22.69	
	2	9025	215.037744	52.913979	24.41	2	9097	214.979175	52.949327	24.22	
	2	9063	214.970329	52.914055	24.27	2	9137	214.990387	52.916955	23.96	
	2	9090	215.008556	52.948859	24.72	2	9142	214.978554	52.921541	24.03	
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	215.01005916666 668 Degrees 52.92765 Degrees	23.770644252491 728			3	18	26522.602
	2	1 (G140M/F100LP)	c1	3 Shutter Slitlet	215.01005916666 668 Degrees 52.92765 Degrees	23.770644252491 728			3	18	26522.602
	3	1 (G140M/F100LP)	c1	3 Shutter Slitlet	215.01005916666 668 Degrees 52.92765 Degrees	23.770644252491 728			3	18	26522.602
	4	1 (G140M/F100LP)	c1	3 Shutter Slitlet	215.01005916666 668 Degrees 52.92765 Degrees	23.770644252491 728			3	18	26522.602
	5	2 (G235M/F170LP)	c1	3 Shutter Slitlet	215.01005916666 668 Degrees 52.92765 Degrees	23.770644252491 728			3	30	28886.002
	6	3 (G395M/F290LP)	c1	3 Shutter Slitlet	215.01005916666 668 Degrees 52.92765 Degrees	23.770644252491 728			3	9	10635.301
Special Requirements	Group Visits within 53.0 Days Visits Same PA MSA Scheduled Aperture PA 23.7566 to 23.7566 Degrees (V3 245.18202 to 245.18202)										

Proposal 9214 - Observation 9 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Observation	Proposal 9214, Observation 9: A2744_replan				Tue Jun 16 21:00:52 GMT 2026
	Diagnostic Status: Warning				
Observing Template: NIRSpec MultiObject Spectroscopy					
Diagnostics	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#1) has 11 primary slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#1) has 16 filler slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#1) has 4 filler slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#1) has 4 primary slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#2) has 11 primary slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#2) has 16 filler slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#2) has 4 filler slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#2) has 4 primary slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#3) has 11 primary slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#3) has 16 filler slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#3) has 4 filler slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#3) has 4 primary slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#4) has 11 primary slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#4) has 16 filler slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#4) has 4 filler slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#4) has 4 primary slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#5) has 11 primary slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#5) has 16 filler slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#5) has 4 filler slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#5) has 4 primary slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#6) has 11 primary slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#6) has 16 filler slit traces affected by failed open shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#6) has 4 filler slits affected by failed closed shutters.				
	(A2744_replan (Obs 9)) Warning (Form): Config c1 (#6) has 4 primary slits affected by failed closed shutters.				
	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	(Visit 9:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
	Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections
(6)		A2744_v4	RA: 00 14 22.3021 (3.5929254d) Dec: -30 24 9.80 (-30.40272d) Equinox: J2000		
Comments: Description=[]					

Proposal 9214 - Observation 9 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
	2	Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	fixed (55 sources)	A2744_v4 (4491 sources)	jwst-nirspec-mr	1.5			
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	4343	3.603004	-30.430964	23.95	1	4405	3.604209	-30.385654	22.36	
	1	4364	3.574588	-30.415319	23.56	1	4476	3.584309	-30.412459	24.5	
	1	4390	3.574002	-30.394799	23.89	1	4479	3.571667	-30.379687	23.43	
	1	4394	3.576655	-30.391370	23.84	1	4491	3.584418	-30.383041	23.07	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	4343	3.603004	-30.430964	23.95	2	4405	3.604209	-30.385654	22.36	
	2	4364	3.574588	-30.415319	23.56	2	4476	3.584309	-30.412459	24.5	
	2	4390	3.574002	-30.394799	23.89	2	4479	3.571667	-30.379687	23.43	
	2	4394	3.576655	-30.391370	23.84	2	4491	3.584418	-30.383041	23.07	
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.595240125 Degrees - 30.402363055555 554 Degrees	182.20916088616 215			3	18	26522.602
	2	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.595240125 Degrees - 30.402363055555 554 Degrees	182.20916088616 215			3	18	26522.602
	3	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.595240125 Degrees - 30.402363055555 554 Degrees	182.20916088616 215			3	18	26522.602
	4	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.595240125 Degrees - 30.402363055555 554 Degrees	182.20916088616 215			3	18	26522.602
	5	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.595240125 Degrees - 30.402363055555 554 Degrees	182.20916088616 215			3	30	28886.002
	6	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.595240125 Degrees - 30.402363055555 554 Degrees	182.20916088616 215			3	9	10635.301

Proposal 9214 - Observation 9 - Unlocking the nature of the first galaxies with ultra-deep rest-UV spectroscopy

Special Requirements	Group Visits within 53.0 Days Visits Same PA MSA Scheduled Aperture PA 182.2103 to 182.2103 Degrees (V3 43.63576 to 43.63576)
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